

2020

Linear Algebra and Calculus Final Exam

- ① R is the region bounded by $f(x) = 9 - (\frac{x}{2})^2$ and $g(x) = 6 - x$, find the area of R .

We want to compute $\int_R (f(x) - g(x)) dx$, so we first need to find the integration limits, where f and g cut, so $f(x) = g(x)$

$$\rightarrow f(x) - g(x) = 0 \rightarrow 9 - \frac{x^2}{4} - 6 + x = 0 \rightarrow -\frac{x^2}{4} + x + 3 = 0 \rightarrow x = \frac{-1 \pm \sqrt{1+3}}{-\frac{1}{2}} = \frac{-1 \pm 2}{-\frac{1}{2}} = 2 \pm 4 \rightarrow \begin{matrix} x=6 \\ x=-2 \end{matrix} \text{ Now}$$

$$\int_R (f(x) - g(x)) dx = \int_{-2}^6 -\frac{x^2}{4} + x + 3 dx = \left[-\frac{x^3}{12} + \frac{x^2}{2} + 3x \right]_{-2}^6 = -18 + 18 + 18 - \frac{2}{3} - 2 + 6 = 22 - \frac{2}{3} = \frac{66-2}{3} = \frac{64}{3}$$

- ② Compute the slope of the tangent line to the function $f(x) = \sqrt{1 + \cos 2x}$ at $x = \frac{\pi}{4}$.

Such slope is just $f'(\frac{\pi}{4})$. So, $f'(x) = \frac{1}{2} \frac{1}{\sqrt{1 + \cos 2x}} \cdot (-\sin 2x) \cdot 2$ by the Chain rule. So $f'(x) = \frac{-\sin(2x)}{\sqrt{1 + \cos 2x}}$

$$\text{Now } f'(\frac{\pi}{4}) = -1.$$

- ③ $f(x) = \begin{cases} 3x-2 & x < 2 \\ g(x) & x = 2 \\ x^2 & x > 2 \end{cases}$, find $g(x)$ such that $f(x)$ is continuous.

For f to be continuous, we have to make sure that $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = f(2)$, since $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (3x-2) = 4$ and $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} x^2 = 4$, so we need a function $g(x)$ such that $g(2) = 4$.

For example $g(x) = 2x$.

- ④ Does $\lim_{x \rightarrow \frac{\pi}{2}} \tan x$ exist?

For that, the lateral limits should coincide, so $\lim_{x \rightarrow \frac{\pi}{2}^-} \tan x = \lim_{x \rightarrow \frac{\pi}{2}^-} \frac{\sin x}{\cos x} = +\infty$, since on the left of $\frac{\pi}{2}$ the cosine is positive (and sin is 1) and approaches 0.

Now, $\lim_{x \rightarrow \frac{\pi}{2}^+} \tan x = \lim_{x \rightarrow \frac{\pi}{2}^+} \frac{\sin x}{\cos x} = -\infty$, since the cosine on the right is negative while approaching 0. Since the lateral limits don't coincide, the limit doesn't exist.

- ⑤ $\iint_R y + x^2 dA$, where $R = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1, y \geq 0, y \leq x\}$

$$\text{This means } \iint_R y + x^2 dA = \int_0^1 \int_0^x y + x^2 dy dx = \int_0^1 \left[\frac{y^2}{2} + x^2 y \right]_{y=0}^{y=x} dx = \int_0^1 \frac{x^2}{2} + x^3 dx = \left[\frac{x^3}{6} + \frac{x^4}{4} \right]_0^1 = \frac{1}{6} + \frac{1}{4} = \frac{5}{12}$$

- ⑥ Let $A = \begin{bmatrix} 1 & 2 & 0 & -1 & 3 \\ 0 & 1 & 2 & 1 & 4 \\ 1 & 3 & 2 & 0 & 7 \end{bmatrix}$.

a) Can $Ax = 0$ have a unique solution?

No, since A has more columns than rows and therefore free variables.

b) Basis for $\text{Nul } A$.

We need to find the solutions for $Ax=0$ by row reducing $[A|0]$.

$$\left[\begin{array}{ccccc|c} 1 & 2 & 0 & -1 & 3 & 0 \\ 0 & 1 & 2 & 1 & 4 & 0 \\ 1 & 3 & 2 & 0 & 7 & 0 \end{array} \right] \sim \left[\begin{array}{ccccc|c} 1 & 2 & 0 & -1 & 3 & 0 \\ 0 & 1 & 2 & 1 & 4 & 0 \\ 0 & 1 & 2 & 1 & 4 & 0 \end{array} \right] \sim \left[\begin{array}{ccccc|c} 1 & 0 & -4 & -3 & -5 & 0 \\ 0 & 1 & 2 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \begin{cases} x_1 = 4x_3 + 3x_4 + 5x_5 \\ x_2 = -2x_3 - x_4 - 4x_5 \end{cases}$$

$$\rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = x_3 \begin{bmatrix} 4 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 3 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 5 \\ -4 \\ 0 \\ 0 \\ 1 \end{bmatrix} \text{ So a basis is } \left\{ \begin{bmatrix} 4 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ -4 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

c) Find a basis for $\text{Col } A$.

The basis will be made out of the pivot columns of A , which, by the reduction in part b), are the first two columns, so a basis for $\text{Col } A$ is $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \right\}$.

⑦ Let A be any square matrix.

a) Show that if $A^2 - 3A + I = 0 \rightarrow A^{-1} = 3I - A$.

$A^2 - 3A + I = 0 \rightarrow A^2 - 3A = -I \rightarrow -A^2 + 3A = I$, taking common factor on the right we get $(-A + 3I)A = I$. since there is a matrix multiplying A that gives the identity, it must be the inverse, so $A^{-1} = 3I - A$.

b) Show if $a_{ij} = 0$ for $i \neq j$ and $a_{ii} \neq 0$ for $i = j$, then A is invertible.

This means that $A = \begin{bmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{bmatrix}$, since all the elements in the diagonal are different from 0,

$\det A = a_{11} \dots a_{nn} \neq 0 \rightarrow A$ is invertible.

c) Find A^{-1} .

The inverse of a diagonal matrix is a diagonal with the inverses of the entries, that is $A^{-1} = \begin{bmatrix} \frac{1}{a_{11}} & 0 & \dots & 0 \\ 0 & \frac{1}{a_{22}} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \frac{1}{a_{nn}} \end{bmatrix}$.

⑧ Let V be the vector space of functions $f: \mathbb{R} \rightarrow \mathbb{R}$, and $C^1(\mathbb{R})$ the subset of V of all the functions that are differentiable. Show that $C^1(\mathbb{R})$ is a subspace of V .

For that we need to prove 3 things:

- $0 \in C^1(\mathbb{R})$, indeed 0 is the constant function $f(x) = 0$, which is differentiable, so $f(x) = 0 \in C^1(\mathbb{R})$.
- If $f, g \in C^1(\mathbb{R}) \rightarrow f+g \in C^1(\mathbb{R})$. Indeed if f and g are differentiable, then the sum is also differentiable, $f+g \in C^1(\mathbb{R})$.
- If $f \in C^1(\mathbb{R})$ and $k \in \mathbb{R} \rightarrow kf \in C^1(\mathbb{R})$. Indeed if f is differentiable, $(kf)(x) = k \cdot f(x)$ is also differentiable, $kf \in C^1(\mathbb{R})$.

⑨ Let A be a square matrix.

a) Show that if A diagonalizable $\rightarrow A^k = PD^kP^{-1}$.

$A = PDP^{-1} \rightarrow A^2 = (PDP^{-1})^2 = (PDP^{-1})(PDP^{-1}) = P \underbrace{D^{-1}D}_{I} P^{-1} = PD^2P^{-1}$. Applying this k times gives $A^k = PD^kP^{-1}$.

b) $A = \begin{bmatrix} 1 & -6 \\ 2 & -6 \end{bmatrix}$. Find a formula for A^n .

First we need to diagonalize A (if possible). Start by computing the eigenvalues, by solving $\det(A - \lambda I) = 0$.

$$\det(A - \lambda I) = (1 - \lambda)(-6 - \lambda) + 12 = -6 - \lambda + 6\lambda + \lambda^2 + 12 = \lambda^2 + 5\lambda + 6 = 0 \rightarrow \lambda = \frac{-5 \pm \sqrt{25 - 24}}{2} = \frac{-5 \pm 1}{2} \begin{cases} \lambda = -2 \\ \lambda = -3 \end{cases}$$

Since we have two different eigenvalues, the matrix diagonalizes. Let's find the eigenvectors by solving $(A - \lambda I)x = 0$.

$$\lambda = -2$$

$$A + 2I = \begin{bmatrix} 3 & -6 \\ 2 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix} \rightarrow x_1 = 2x_2 \rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \text{So } D = \begin{bmatrix} -2 & 0 \\ 0 & -3 \end{bmatrix}, P = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \rightarrow P^{-1} = \frac{1}{1} \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$$

$$\lambda = -3$$

$$A + 3I = \begin{bmatrix} 4 & -6 \\ 2 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & -3/2 \\ 0 & 0 \end{bmatrix} \rightarrow x_1 = \frac{3}{2}x_2 \rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \text{Therefore } A^n = P D^n P^{-1} = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} (-2)^n & 0 \\ 0 & (-3)^n \end{bmatrix} \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix} =$$

$$= \begin{bmatrix} 2(-2)^n & 3(-3)^n \\ (-2)^n & 2(-3)^n \end{bmatrix} \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix} =$$

$$= \begin{bmatrix} 4(-2)^n - 3(-3)^n & -6(-2)^n + 6(-3)^n \\ 2(-2)^n - 2(-3)^n & -3(-2)^n + 4(-3)^n \end{bmatrix}$$

10) Let $w_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $w_2 = \begin{bmatrix} 3 \\ -1 \\ -1 \end{bmatrix}$ and $y = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$, and let $W = \text{Span}\{w_1, w_2\}$.

a) Find a basis for W of orthogonal vectors.

Since w_1 and w_2 are linearly independent, we can apply the Gram-Schmidt process to get $\{v_1, v_2\}$.

$$v_1 = w_1.$$

$$v_2 = w_2 - \frac{w_2 \cdot v_1}{v_1 \cdot v_1} v_1 = \begin{bmatrix} 3 \\ -1 \\ -1 \end{bmatrix} - \frac{4}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ -2 \end{bmatrix}.$$

b) Express y as the sum of a vector in W and one in $W^\perp$.

By the Orthogonal Decomposition Theorem, we need to find the orthogonal projection of y onto W , that is

$$\text{proj}_W y = \frac{y \cdot v_1}{v_1 \cdot v_1} v_1 + \frac{y \cdot v_2}{v_2 \cdot v_2} v_2 = \frac{8}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \frac{8}{16} \begin{bmatrix} 2 \\ -2 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} \in W, \text{ now for the vector } z \text{ in } W^\perp, z = y - \text{proj}_W y =$$

$$= \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ -1 \end{bmatrix}.$$

11) True or false

a) $T(x) = (x_2, x_3, x_1^2)$. Since $T(0) = 0$, T is linear.

False, $T(3x) = (3x_2, 3x_3, 9x_1^2) \neq (3x_2, 3x_3, 3x_1^2) = 3T(x)$.

b) $V = \mathbb{R}^2$, $B = \{b_1, b_2\}$, $C = \{c_1, c_2\}$, then if $[c_1, c_2, b_1, b_2] \sim [I, P]$ means $[x]_B = P[x]_C \quad \forall x \in V$.

False, it's $[x]_C = P[x]_B$.

c) If $\{u_1, \dots, u_p\}$ is an orthonormal basis for a subspace W of $\mathbb{R}^n$, then $\text{proj}_W y = (y \cdot u_1)u_1 + \dots + (y \cdot u_p)u_p$

True, since the complete formula is $\text{proj}_W y = \frac{(y \cdot u_1)}{u_1 \cdot u_1} u_1 + \dots + \frac{(y \cdot u_p)}{u_p \cdot u_p} u_p$, and the basis is orthonormal, each denominator is the dot product of unit vectors, so $u_i \cdot u_i = 1$.